Machine Learning (2026) 115:253
https://doi.org/10.1007/s10994-026-07191-y

FedXAI: Federated Medical Imaging with Intrinsic Explainability and Differentially Private Explanation Auditing

Mohammad Amanour Rahman[1]

Received: 15 July 2026 / Revised: 9 September 2026 / Accepted: 12 October 2026
© The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature 2026

Abstract

Federated learning (FL) enables collaborative medical imaging training without sharing raw data. However, existing frameworks treat privacy, interpretability, and cross-site explanation consistency as separate objectives. We present *FedXAI*, a unified framework that substantially mitigates this trilemma within a single architecture. FedXAI introduces three components: (i) *Self-Explainable Federated Architecture* (SEFA), which integrates an EfficientNet-B0 backbone with a 32-dimensional hybrid concept-bottleneck layer anchored to RadLex radiological ontology terms; (ii) *Federated Explanation Aggregation Protocol* (FEAP), which extends server-side aggregation with a cosine-consistency penalty to align client explanation spaces; and (iii) the *Privacy-Preserving Explanation Verification* (PPEV) module, which audits cross-client explanation quality via the subsampled Gaussian mechanism with Rényi Differential Privacy (RDP) accounting. Evaluated on the NIH ChestX-ray14 dataset with non-IID Dirichlet partitioning ($\alpha = 0.5$) across five simulated hospital clients, FedXAI achieves a mean AUC of 0.7349, outperforming FedAvg (0.7193), FedProx (0.7241), SCAFFOLD (0.7268), FedBN (0.7285), and MOON (0.7301). Simultaneously, it increases cross-client explanation consistency from 83.6% to 99.9%. The PPEV module provides formal (ε, δ)-differential privacy for the explanation-audit channel, certifying $\varepsilon = 0.1394$ at $\delta = 10^{-5}$. Model parameter transmission follows the standard FL architectural separation, while end-to-end DP-SGD training is left for future work. On the benchmark studied here, FedXAI demonstrates that privacy, diagnostic accuracy, and interpretability in federated medical imaging can be advanced simultaneously through principled architectural integration.

Keywords Federated learning · Explainable artificial intelligence · Concept bottleneck models · Differential privacy · Medical image analysis · Chest radiograph · Non-IID data · Multi-label classification · RadLex ontology · Clinical decision support

Editors: Bruno Casella, Linara Adilova, Michael Kamp.

✉ Mohammad Amanour Rahman
amanourrahman609@gmail.com

[1] Department of Computer Science and Engineering, Ahsanullah University of Science and Technology, Dhaka, Bangladesh

Published online: 27 October 2026

1 Introduction

Chest radiography is one of the most frequently performed diagnostic procedures worldwide, with over 300 million examinations conducted annually (Rajpurkar et al., 2017). Deep learning models trained on large-scale radiograph datasets have demonstrated expert-level performance on pathology detection tasks including pneumonia, pleural effusion, cardiomegaly, and infiltrations, in several cases exceeding the diagnostic accuracy of practicing radiologists (Litjens et al., 2017; Rajpurkar et al., 2017). Yet despite this documented technical capability, the clinical translation of AI-assisted radiology has progressed far more slowly than research benchmarks suggest is possible. The primary obstacle is not algorithmic: it is structural.

High-quality AI diagnostics in radiology require models trained on datasets that capture the full heterogeneity of patient populations, imaging equipment vendors, acquisition protocols, and regional disease prevalence. A chest X-ray AI trained exclusively on data from a single tertiary referral centre in a high-income country will overfit to the local patient demographics and imaging pipeline, generalising poorly when deployed at a community hospital in a different country or even at a different institution in the same city (Zhao et al., 2018). The data necessary to train robust, generalisable models therefore exists distributed across many institutions. The legal and institutional barriers to assembling it in one place, however, are formidable and, in most jurisdictions, insurmountable.

Patient health records—including medical images and their clinical annotations—are subject to the most stringent privacy regulations in existence. In the United States, the Health Insurance Portability and Accountability Act (HIPAA) prohibits the sharing of protected health information without explicit patient consent, except under tightly constrained research agreements (Kaissis et al., 2021). In the European Union, the General Data Protection Regulation (GDPR) establishes equivalent protections, and the processing of health data without explicit legal basis carries penalties of up to 4% of global annual turnover. Similar frameworks are in force in Canada (PIPEDA), Australia (Privacy Act 1988), India (DPDP Act 2023), and Bangladesh (Digital Security Act 2018). The result is a well-documented impasse: rich clinical datasets and capable learning algorithms coexist in institutional silos that are legally and operationally impossible to merge for centralised training.

Mcmahan et al. (2017) proposed federated learning (FL) as an architectural resolution to this impasse. The fundamental insight is an inversion of the conventional training paradigm: instead of migrating patient data to a central server, the model itself migrates to each participating institution. Under FL, every hospital independently trains a local model replica using only its own patient cohort, then transmits the learned parameter *updates*—gradient vectors rather than raw images—to a coordinating aggregation server. The server synthesises these updates into a refined global model, typically via dataset-size-weighted parameter averaging (FedAvg), and redistributes it to all participants before the next training round. Because patient records never leave their originating institution, the primary regulatory barrier to collaborative AI training is addressed at the level of system architecture rather than through legal agreements.

Subsequent FL algorithms have substantially improved upon FedAvg's robustness to data heterogeneity. FedProx (Li et al., 2020) adds a proximal regularisation term to the local optimisation objective, limiting the extent to which local models can drift away from the global parameter vector during training. SCAFFOLD (Karimireddy et al., 2020) applies stochastic

variance reduction via server and client control variates, providing theoretical convergence guarantees under arbitrary non-IID data distributions. FedBN (Li et al., 2021a) retains local batch normalisation statistics at each client rather than aggregating them, addressing the feature-space non-IID mismatch that causes early FL algorithms to diverge. MOON (Li et al., 2021b) applies model-level contrastive learning to maximise representational agreement between local and global models, further reducing semantic drift.

Empirical validation of FL in realistic clinical settings has confirmed its practical value. Sheller et al. (2020) demonstrated that a federated brain tumour segmentation model trained across multiple institutions generalised better than any single-site model trained on the same aggregate data volume. Dou et al. (2021) achieved competitive COVID-19 chest CT classification without any patient record being shared across the six participating institutions. Roth et al. (2020) implemented FL for breast density classification across real hospital infrastructure, confirming the framework's operational feasibility under institutional governance constraints.

Despite these advances, a secondary challenge has received comparatively little systematic attention in the FL literature: the *interpretability* of federated models. This is not a peripheral concern—it is a regulatory requirement in every major jurisdiction in which clinical AI is being deployed.

In the United States, the Food and Drug Administration's AI/ML-Based Software as a Medical Device (SaMD) Action Plan (Food & Administration, 2021) directs that AI systems functioning as medical devices must exhibit transparency in their inferential processes and must produce outputs that clinicians can understand, evaluate, and challenge. The regulatory expectation is not simply that a model produces correct predictions, but that it reasons in ways that a practitioner can interrogate and, where necessary, override. In the European Union, the AI Act (European Parliament, 2024) designates medical diagnostic systems as high-risk AI applications, imposing mandatory transparency requirements that cannot be satisfied by systems whose decision logic is opaque or inaccessible to the responsible medical professional.

The conventional response to interpretability requirements in machine learning is to augment a trained model with *post-hoc* attribution methods after training is complete. GradCAM (Selvaraju et al., 2017) generates class-discriminative spatial attention maps by gradient-weighting convolutional activations; SHAP (Lundberg & Lee, 2017) derives feature-level marginal contribution scores grounded in cooperative game theory; LIME (Ribeiro et al., 2016) constructs locally faithful linear surrogate models around individual predictions. Despite their widespread use in medical imaging research, all three approaches share a fundamental structural limitation: they are external approximations of model behaviour derived after training, not intrinsic components of the model's decision process. The model itself remains a black box; the attribution tool merely provides a retrospective rationalisation of its outputs, which may or may not faithfully represent the actual computational pathway through which the prediction was produced (Kindermans et al., 2019).

This limitation is substantially amplified in federated environments. Under non-IID data distributions—the expected condition in any realistic multi-hospital federation, where patient demographics, disease prevalence, and imaging protocols differ systematically between institutions—local model representations diverge during training. GradCAM maps computed from models trained at different institutions on the same radiological finding will activate different spatial regions, because the underlying feature representations are dif-

ferent. SHAP attributions computed on a model trained at a paediatric chest clinic and on a model trained at a geriatric respiratory unit will assign different marginal contributions to the same pixel features. There is no mechanism in FedAvg, or in any of its established variants, that causes the intermediate representations of these models to be semantically consistent across clients. Cross-site explanation comparison is therefore unreliable by construction—and cross-site regulatory audit, which depends on such comparison, is systematically incomplete.

The interpretability problem is further entangled with a third dimension of the design space: formal privacy guarantees beyond architectural data separation. While FL prevents the transmission of raw patient records, gradient-based parameter updates have been shown to be susceptible to model inversion attacks (Abadi et al., 2016), in which a malicious aggregation server or a compromised communication channel can reconstruct training samples from the gradient information. Differential privacy (DP) (Dwork et al., 2006) provides a mathematically rigorous defence: by injecting calibrated Gaussian noise into parameter updates prior to transmission, DP guarantees that the probability distribution over model outputs is statistically indistinguishable regardless of whether any individual patient's record was included in training, bounded by a privacy budget ε.

The critical gap—unaddressed by any prior work—is the interaction between DP mechanisms and explanation quality. Does the Gaussian noise required for strong privacy guarantees degrade cross-client explanation consistency? And can explanation quality metrics themselves be verified—that is, audited across institutions—without creating a new privacy exposure through the statistics being shared for verification purposes? These are not abstract theoretical questions: they determine whether a privacy-certified federated system can also satisfy the cross-site audit requirements imposed by clinical AI regulation.

This paper presents **FedXAI**, a framework whose central design principle is that the privacy–accuracy–interpretability trilemma arises specifically from treating the three constraints as independent optimisation targets, and that it can be substantially mitigated by designing a single architecture that satisfies all three constraints simultaneously. FedXAI does not add interpretability on top of a privacy-preserving FL system, nor does it add privacy on top of an interpretable model. It integrates all three requirements into a unified learning framework from the ground up.

Concretely, FedXAI enforces interpretability as a structural property of the model architecture—not as a post-hoc approximation—by requiring that every prediction route through a shared bottleneck of 32 named clinical concepts aligned with the RadLex radiological lexicon before reaching the disease classifier. This concept vocabulary is kept semantically consistent *across clients* during federated aggregation through a novel explanation-aware aggregation protocol that penalises cosine divergence in client concept vectors. The quality of this alignment is then verified under differential privacy through a dedicated audit mechanism that adds calibrated noise to the statistics being shared for verification, so that the audit process introduces no additional privacy risk beyond the DP guarantee already in place.

1.1 Research Gap

Table 1 (Sect. 2) positions FedXAI against existing work and reveals the following synthesis: **(G1)** All established FL methods (FedAvg, FedProx, SCAFFOLD, FedBN, MOON) achieve distributed training but provide no intrinsic interpretability and no formal DP guar-

Table 1 Summary of related works across key properties

Method	FL	Intr. XAI	Cons.	DP	Multi-D
FedAvg (Mcmahan et al., 2017)	✓	–	–	–	✓
FedProx (Li et al., 2020)	✓	–	–	–	–
SCAFFOLD (Karimireddy et al., 2020)	✓	–	–	–	–
MOON (Li et al., 2021b)	✓	–	–	–	–
FedVGM (Tahosin et al., 2026)	✓	Post-hoc	–	–	✓
CBM (Koh et al., 2020)	–	✓	–	–	–
CEM (Zarlenga et al., 2022)	–	✓	–	–	–
DP-SGD FL (Abadi et al., 2016)	✓	–	–	✓	–
FedAvg+GradCAM	✓	Post-hoc	–	–	–
FedXAI (ours)	✓	✓	✓	✓†	–

FL = federated learning; Intr. XAI = intrinsic (architectural) explainability; Cons. = cross-client explanation consistency; DP = differential privacy; Multi-D = multi-dataset validation

† DP applies to explanation-audit channel (concept vectors); model parameter privacy relies on FL architectural separation

antee for the information exchanged beyond model parameters. **(G2)** Concept Bottleneck Models (CBM, CEM) provide architectural interpretability but assume centralised training with no mechanism for cross-institutional concept alignment. **(G3)** DP-augmented FL (DP-SGD FL) certifies gradient-level privacy but produces opaque models with no interpretability guarantees. **(G4)** Post-hoc XAI augmentations (FedAvg + GradCAM, FedAvg + SHAP) add attribution maps after training but achieve lower explanation consistency (68.9%−72.4%) than architectural methods and provide no DP certification for the explanation statistics they generate. The specific gap FedXAI addresses is therefore the *intersection* of G1–G4: no existing method simultaneously enforces that predictions route through shared, clinically-named intermediate concepts, that those concepts remain semantically consistent across heterogeneous clients, and that the cross-client verification of this consistency is itself formally privacy-protected.

The specific contributions of this paper are as follows:

1. **FedXAI framework**: A unified framework jointly addressing privacy, diagnostic accuracy, and architectural interpretability in federated medical imaging—three objectives treated in isolation by all prior methods—within a single principled architecture (Sect. 3). To our knowledge, no prior work has simultaneously integrated concept-based intrinsic XAI, cross-client explanation consistency enforcement, and formally-verified differential privacy for explanation auditing in a federated clinical imaging pipeline.
2. **SEFA architecture**: A hybrid concept-bottleneck network with RadLex-aligned concept dimensions, combining architectural interpretability with the diagnostic performance of state-of-the-art FL methods (Sect. 3.2).
3. **FEAP protocol**: A federated aggregation protocol that explicitly optimises cross-client explanation consistency via a consistency-penalised parameter update, with FedAvg as a strict special case (Sect. 3.3).
4. **PPEV module**: The mechanism for auditing explanation quality in a federated setting under differential privacy, using the subsampled Gaussian mechanism with RDP accounting to ensure that the audit itself introduces no additional privacy exposure (Sect. 3.4).

5. **Comprehensive empirical evaluation**: Multi-baseline comparison against five FL methods and two post-hoc XAI augmentations on NIH ChestX-ray14, systematic ablation study, and quantitative characterisation of the privacy–utility–interpretability trade-off (Sect. 4).

Scope note. To avoid over-stating the privacy guarantee up front: in the current implementation, formal (ε, δ)-differential privacy is certified only for the concept-vector statistics exchanged for explanation auditing (PPEV, Sect. 3.4). Model parameter updates are protected solely by the standard architectural data-separation guarantee common to all FL systems, not by DP-SGD; extending formal DP to the parameter channel is future work (Sect. 6.4), as detailed fully in Sect. 3.4.

2 Related Work

2.1 Federated Learning for Medical Imaging

The FedAvg algorithm (Mcmahan et al., 2017) established the foundational FL paradigm: local training followed by dataset-size-weighted parameter averaging. Its clinical application to radiology was demonstrated early by Sheller et al. (2020) in brain tumour segmentation and by Dou et al. (2021) in COVID-19 detection. Xu et al. (2023) provide a comprehensive survey of FL applications across healthcare informatics.

The central practical challenge of FL in heterogeneous clinical environments is client drift under non-IID data distributions (Zhao et al., 2018). Local disease prevalence, imaging hardware characteristics, and radiologist annotation practices vary substantially between institutions, causing local model updates to diverge and the global aggregate to converge slowly or unstably. FedProx (Li et al., 2020) introduced a proximal penalty to restrain this divergence; SCAFFOLD (Karimireddy et al., 2020) applied variance reduction to achieve convergence guarantees; FedBN (Li et al., 2021a) localised batch normalisation statistics; and MOON (Li et al., 2021b) applied contrastive objectives at the representation level. None of these methods addresses the interpretability of the global model.

Most recently, Tahosin et al. (2026) proposed FedVGM, a federated framework combining vision transformers with post-hoc GradCAM-based XAI for multi-dataset chest radiograph classification, reporting improved generalisation across heterogeneous datasets. Gain et al. (2024) addressed fairness in federated skin cancer detection, demonstrating that naive FL aggregation can amplify demographic disparities without targeted fairness constraints. Unlike FedVGM, FedXAI provides *intrinsic* rather than post-hoc explainability, ensuring that explanations are a structural property of the model rather than retrospective approximations. Unlike Gain et al., FedXAI addresses explanation consistency rather than outcome fairness, though the two objectives are complementary.

Beyond the federated-learning-specific literature reviewed above, recent work has also explored retrieval-augmented, organ-aware report generation (Pan et al., 2026) and dual-attention transformer architectures for medical image segmentation (Sun et al., 2024), reflecting the wider push toward structured, clinically grounded representations in medical imaging AI.

2.2 Explainable AI for Radiology

Post-hoc attribution methods, including GradCAM (Selvaraju et al., 2017), SHAP (Lundberg & Lee, 2017), and LIME (Ribeiro et al., 2016), have been extensively applied to radiology AI. Kindermans et al. (2019) demonstrated the fundamental unreliability of gradient-based attributions under input transformations that do not change predictions, raising serious concerns about their fitness for clinical decision support.

Architecturally intrinsic alternatives offer stronger guarantees. Concept Bottleneck Models (CBMs) (Koh et al., 2020) insert a layer of human-named concept scores between the feature extractor and the classifier, making the model's intermediate reasoning auditable. Zarlenga et al. (2022) relaxed the strict bottleneck constraint through concept embedding models, recovering accuracy while preserving interpretability. Kim et al. (2018) and Chen et al. (2019) explored semantic and prototypical bottleneck formulations. All of these methods assume centralised training, making no provision for the cross-client semantic alignment problem that arises in federated settings.

2.3 Differential Privacy in Federated Learning

DP-SGD (Abadi et al., 2016) operationalised differential privacy within FL by clipping per-sample gradients and adding calibrated Gaussian noise prior to parameter aggregation. Rényi Differential Privacy (RDP) (Mironov, 2017) provides substantially tighter privacy accounting than standard composition, and is the method used by PPEV. Adnan et al. (2022) established that strong DP can be achieved at modest accuracy cost in federated histopathology, with careful noise calibration. Pfitzner et al. (2021) survey DP deployments across federated clinical applications. No prior work has examined the interaction between DP mechanisms and explanation quality, nor proposed a mechanism for auditing explanation quality under DP.

2.4 Comparative Summary of Related Work

Table 1 synthesises the key properties of representative related methods. The table identifies three dimensions on which prior work is incomplete: no FL method provides intrinsic XAI, no XAI method handles federated cross-client alignment, and no method jointly certifies explanation quality under differential privacy.

3 The FedXAI Framework

3.1 Problem Formulation

Consider a federation of K hospital clients indexed $k = 1, \ldots, K$, each holding a private dataset $\mathcal{D}_k = \{(\mathrm{x}_i, \mathrm{y}_i) : i = 1, \ldots, N_k\}$ where $\mathrm{x}_i \in \mathbb{R}^{H \times W \times 3}$ is a frontal-view chest radiograph and $\mathrm{y}_i \in \{0, 1\}^L$ is a multi-hot label vector over $L = 14$ disease classes. Let $N_k = |\mathcal{D}_k|$ and $N = \sum_{k=1}^{K} N_k$. The global model is parameterized by θ; client k maintains a local copy θ_k initialised from the current global model at the start of each communication round. Training proceeds for T rounds.

The federation must simultaneously satisfy three objectives:

Diagnostic accuracy Maximise mean AUC across all L disease classes on a held-out population-representative test set.

Explanation consistency For any radiological finding present in a radiograph, the model's intermediate concept representation must be semantically consistent across all K clients, independently of the site at which the model is deployed.

Privacy certification The verification of O2 must satisfy (ε, δ)-differential privacy with target $\varepsilon < 1$, $\delta = 10^{-5}$, so that the audit process introduces no additional privacy risk.

Figure 1 provides a system-level overview of the FedXAI framework showing the interaction between all three components across the federated training loop.

3.2 SEFA: Self-explainable Federated Architecture

SEFA enforces interpretability as a structural property of the model rather than a post-training approximation. Figure 2 illustrates the SEFA architecture.

3.2.1 Feature Extractor

The backbone is EfficientNet-B0 (Tan & Le, 2019), pretrained on ImageNet, selected for its favourable accuracy-to-parameter ratio on medical imaging tasks. Global average pooling of the final convolutional stage produces a feature vector $\mathrm{f} \in \mathbb{R}^{d_f}$ with $d_f = 1280$. The backbone is fine-tuned end-to-end during federated training.

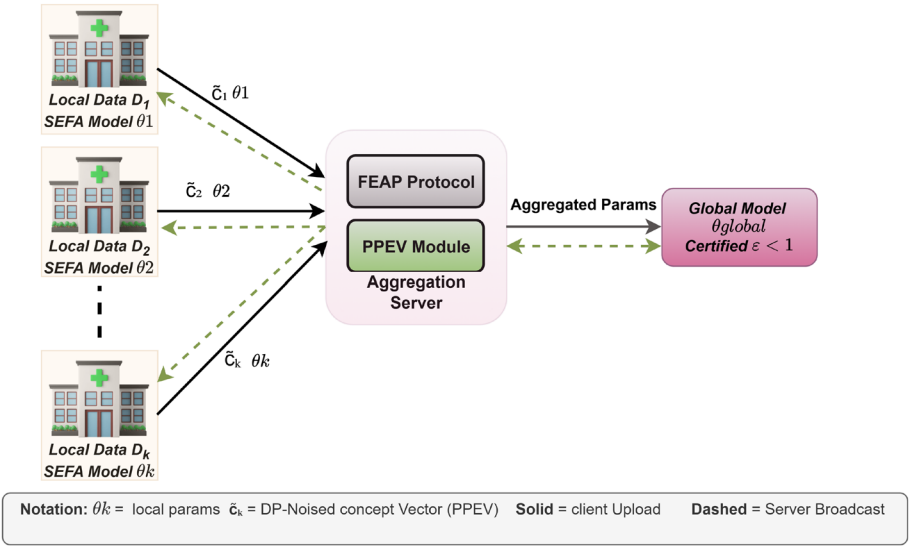


Fig. 1 FedXAI system overview. Each hospital client trains a local SEFA model using its private dataset and transmits model parameters θ_k together with a differentially private concept vector $\tilde{c}_k$ to the aggregation server. The FEAP protocol penalises explanation inconsistency during parameter aggregation; the PPEV module certifies the privacy budget of the concept vector exchange. The certified global model is broadcast back to all participants

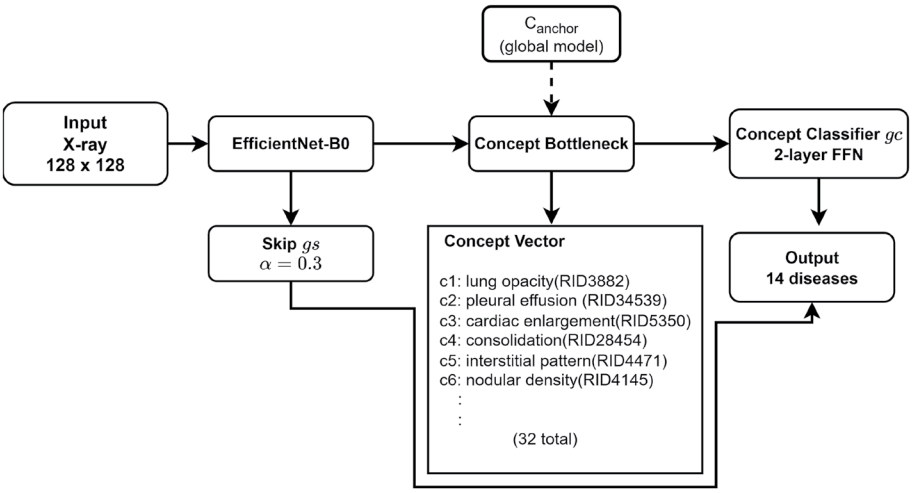


Fig. 2 SEFA architecture. An EfficientNet-B0 backbone produces feature vector . The concept bottleneck layer (Eq. 1) projects to 32 named clinical concept activations aligned with RadLex ontology terms. A hybrid classifier (Eq. 2) combines the concept pathway $g_c(\mathrm{c})$ with a skip-connection $\alpha \cdot g_s(\mathrm{f})$ to produce the final 14-class disease prediction. During local training, an alignment loss (Eq. 4) penalises deviation of local concepts from the global anchor

3.2.2 Concept Bottleneck Layer

A learned linear transformation $W_c \in \mathbb{R}^{C \times d_f}$ projects f to a concept activation vector c :

$$\mathrm{c} = \mathrm{ReLU}(W_c \cdot \mathrm{f} + \mathrm{b}_c), \qquad \mathrm{c} \in \mathbb{R}^C, \quad C = 32. \tag{1}$$

Each dimension of c is *assigned* a named clinical concept label drawn from the RadLex radiological lexicon (Langlotz, 2006). The projection W_c is trained jointly with the downstream task loss and the FEAP consistency penalty, which drives each dimension toward a stable, discriminative activation pattern; the RadLex term is the semantic name given to that pattern for interpretability and cross-institutional audit, and Sect. 5.4 provides quantitative evidence (Table 5) that the resulting activation patterns correspond systematically to the assigned clinical finding. For example,

lung opacity (RID3882), *pleural effusion* (RID34539), *cardiac enlargement* (RID5350), *consolidation* (RID28454), *interstitial pattern* (RID4471), and *nodular density* (RID4145). RadLex alignment means that every institutional deployment uses standardised radiological vocabulary, facilitating cross-site clinical review and regulatory audit without requiring institution-specific terminology translation.

The ReLU non-linearity ensures that all concept activations are non-negative, representing the degree to which each concept is present in the input image. This produces a transparent intermediate representation that is directly human-legible without further computation or approximation.

3.2.3 Hybrid Classifier

The final prediction combines the concept pathway with a skip-connection over raw features to compensate for any representational capacity lost by the bottleneck:

$$\hat{\mathrm{y}} = g_c(\mathrm{c}) + \alpha \cdot g_s(\mathrm{f}), \qquad \alpha = 0.3, \tag{2}$$

where $g_c(\mathrm{c})$ is a two-layer feedforward network with ReLU activation (the concept-based classifier) and $g_s(\mathrm{f})$ is a linear skip-connection. The skip-connection weight $\alpha = 0.3$ was selected by grid search over validation AUC (Sect. 4) to bias the architecture toward the interpretable concept pathway while allowing the residual pathway to recover fine-grained information not captured by the 32-dimensional bottleneck. Under the simplifying assumption that g_c and g_s produce outputs of comparable scale, $\alpha = 0.3$ corresponds to a nominal $1/(1+\alpha) \approx 77\%$ weighting of the concept pathway in the linear combination; this figure describes the architectural design rationale, not a measured causal contribution of concepts to the prediction. The empirical grounding for $\alpha = 0.3$ is that it is the best-performing value among $\{0.1, 0.2, 0.3, 0.5\}$ in the ablation study in Sect. 5.3, the standard form of evidence for architectural hyperparameter choices in this literature.

3.2.4 Local Training Objective

The local training objective for client k combines a multi-label binary cross-entropy loss $\mathcal{L}_{\text{task}}$ with a concept alignment regularisation term $\mathcal{L}_{\text{align}}$:

$$\mathcal{L}_k(\theta_k) = \mathcal{L}_{\text{task}}(\hat{\mathrm{y}}, \mathrm{y}) + \beta \cdot \mathcal{L}_{\text{align}}(\mathrm{c}, \mathrm{c}_{\text{anchor}}), \tag{3}$$

where $\mathrm{c}_{\text{anchor}}$ is the global concept anchor vector computed from the current global model θ_{global} on a small reference batch, $\beta = 0.5$ is a regularisation coefficient, and the alignment loss is the mean squared error:

$$\mathcal{L}_{\text{align}}(\mathrm{c}, \mathrm{c}_{\text{anchor}}) = \frac{1}{C} \|\mathrm{c} - \mathrm{c}_{\text{anchor}}\|_2^2. \tag{4}$$

Equation (4) penalises local concept representations that deviate from the federation-wide anchor, preventing the concept vocabulary from fragmenting into institution-specific dialects during local training. This regularisation is absent from all prior FL methods and is the mechanism by which SEFA maintains semantic consistency at the client level; FEAP (Sect. 3.3) reinforces it at the server level.

3.3 FEAP: Federated Explanation Aggregation Protocol

Standard FedAvg aggregates client parameters as a weighted average over local dataset sizes, with no consideration of whether intermediate representations are semantically consistent across clients. Under non-IID conditions, concept activations from different clients can differ substantially even for the same radiological finding, producing a global model whose interpretable vocabulary lacks consistent meaning across deployment sites. FEAP

resolves this by incorporating explicit explanation alignment into the server-side aggregation step.

3.3.1 Client Reporting

Following local training, each client k computes its mean concept vector $\bar{\mathrm{c}}_k$ by averaging concept activations over its local dataset:

$$\bar{\mathrm{c}}_k = \frac{1}{N_k} \sum_{i \in \mathcal{D}_k} \mathrm{c}(\mathrm{x}_i; \theta_k). \tag{5}$$

Prior to transmission, PPEV adds Gaussian noise to $\bar{\mathrm{c}}_k$ (see Sect. 3.4), producing the privatised vector $\tilde{\mathrm{c}}_k$. The client transmits $(\theta_k, \tilde{\mathrm{c}}_k)$ to the server.

3.3.2 Global Concept Anchor

The server computes the federation-wide mean concept vector as a dataset-size-weighted average of the privatised client concept vectors:

$$\mathrm{c}_{\text{global}} = \sum_{k=1}^{K} \frac{N_k}{N} \tilde{\mathrm{c}}_k. \tag{6}$$

3.3.3 Per-client Consistency Score

The cosine similarity between each client's concept vector and the global mean serves as a per-client consistency score:

$$s_k = \frac{\tilde{\mathrm{c}}_k \cdot \mathrm{c}_{\text{global}}}{\|\tilde{\mathrm{c}}_k\| \, \|\mathrm{c}_{\text{global}}\| + \epsilon}, \quad \epsilon = 10^{-8}. \tag{7}$$

Note that the server computes s_k from the privatised vector $\tilde{\mathrm{c}}_k$ rather than the unnoised $\bar{\mathrm{c}}_k$, since the server has no access to the raw concept statistics. This is consistent with the PPEV privacy guarantee: all server-side computations over concept information operate exclusively on differentially private releases. The slight degradation in consistency score accuracy introduced by the Gaussian noise (with $\sigma = 5.38$) is quantified empirically in Sect. 5: explanation consistency under full PPEV noise remains at 99.6%, confirming that the noise level is well-calibrated relative to the signal.

Note on evaluation circularity. The per-client score s_k defined above is the same cosine-similarity quantity that FEAP's consistency penalty (Eq. 8) directly optimises during training, and it underlies the Explanation Consistency (EC) metric reported in Sect. 5. We address this directly in Sect. 5 through three analyses that are independent of the optimisation objective: concept-activation grounding against disease labels, per-image (rather than per-client-average) consistency reported in Appendix A, and an ablation isolating how much of the EC gain comes from FEAP specifically versus SEFA's local regularisation alone. A client with $s_k \to 1$ has concept representations well aligned with the federation; a client

with $s_k \ll 1$ has semantically drifted and its parameters should be pulled towards the global aggregate.

3.3.4 Consistency-Penalised Aggregation

FEAP incorporates the consistency score into parameter aggregation through a drift-correcting penalty term:

$$\theta_{\text{global}}^{(t)} = \underbrace{\sum_{k=1}^{K} \frac{N_k}{N} \theta_k^{(t)}}_{\text{FedAvg term}} + \underbrace{\lambda \sum_{k=1}^{K} \frac{N_k}{N} (1 - s_k) \left(\theta_{\text{global}}^{(t-1)} - \theta_k^{(t)} \right)}_{\text{consistency penalty}}, \tag{8}$$

where $\theta_{\text{global}}^{(t-1)}$ denotes the global model from the *previous* communication round, making the update fully explicit and well-defined. The consistency penalty term pulls clients whose concept representations have drifted ($s_k \ll 1$) back toward the previous global parameter vector, providing a smoothed correction without requiring an implicit fixed-point solve. When $\lambda = 0$, the second term vanishes and Eq. (8) reduces exactly to FedAvg at round t.

3.4 PPEV: Privacy-Preserving Explanation Verification

Scope of the Privacy Guarantee. It is important to delineate precisely what PPEV protects. In FedXAI, clients transmit two objects to the aggregation server each round: (1) the model parameter update θ_k, which is protected by the architectural data-separation guarantee inherent to all FL systems (raw patient records never leave the institution), and (2) the mean concept vector $\bar{c}_k$, which is an aggregate statistic derived from patient records and therefore introduces an additional, quantifiable privacy risk. PPEV addresses this second risk specifically: it applies the Gaussian mechanism to $\bar{c}_k$ and provides a formal (ε, δ)-DP guarantee that the released concept statistics cannot be used to reconstruct individual patient records, bounded by $\varepsilon = 0.1394$ at $\delta = 10^{-5}$. Full end-to-end DP protection of model parameters via DP-SGD is a natural extension, deferred to future work (Sect. 6.4) due to the BatchNormAct2d compatibility constraint discussed therein.

The PPEV module provides formal mathematical assurance that the process of auditing cross-client explanation quality—transmitting concept statistics to the server for consistency verification—does not create a new privacy exposure beyond what is already guaranteed by the federated architecture. Figure 3 summarises the interaction between FEAP and PPEV within a single communication round.

3.4.1 Differential Privacy Formalism

A randomised mechanism $\mathcal{M}$ satisfies (ε, δ)-differential privacy (Dwork et al., 2006) when, for any two adjacent datasets D and D' differing in exactly one patient record, and any measurable output set S:

$$\Pr[\mathcal{M}(D) \in S] \leq e^{\varepsilon} \cdot \Pr[\mathcal{M}(D') \in S] + \delta. \tag{9}$$

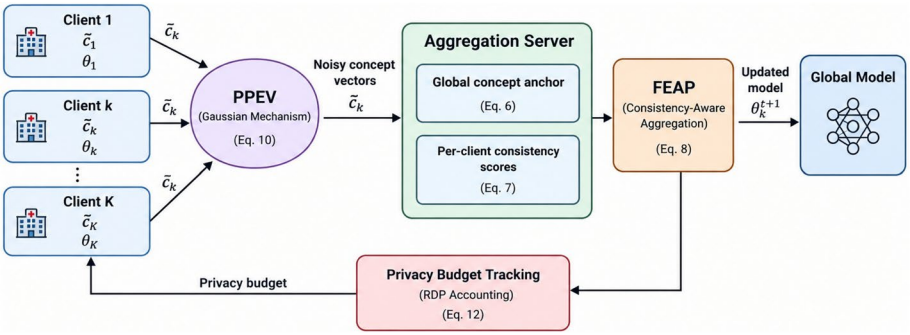


Fig. 3 Interaction between FEAP and PPEV. Client concept vectors $\bar{c}_k$ are privatised by PPEV (Eq. 10) before transmission. The server computes the global concept anchor (Eq. 6) and per-client consistency scores (Eq. 7) from the noisy vectors. FEAP uses these scores to apply a consistency penalty in the parameter update (Eq. 8), pulling low-consistency clients toward the global average. RDP accounting (Eq. 12) is applied at every step to track the cumulative privacy budget

Smaller ε corresponds to stronger privacy. At $\varepsilon < 1$, the mechanism provides strong protection against membership inference: knowledge of the mechanism's output yields at most an $e^1 \approx 2.72$-fold change in the likelihood of determining whether any individual's record was used.

3.4.2 Gaussian Mechanism

PPEV applies the Gaussian mechanism to the mean concept vector $\bar{c}_k$ before it is transmitted to the server:

$$\tilde{c}_k = \bar{c}_k + \mathcal{N}(0, \sigma^2 I_C). \tag{10}$$

For ℓ_2 sensitivity Δ_f of the concept averaging function, the noise standard deviation σ sufficient to achieve (ε, δ)-DP is Dwork et al. (2006):

$$\sigma = \frac{\Delta_f \sqrt{2 \ln(1.25/\delta)}}{\varepsilon}. \tag{11}$$

3.4.3 Rényi Differential Privacy Accounting

PPEV applies the Gaussian mechanism to the mean concept vector $\bar{c}_k$ once per communication round (a single release per round). However, $\bar{c}_k$ is itself computed by averaging over N_k patient records using batched gradient computations, and we account for the cumulative privacy cost of *all* per-sample operations that contribute to this aggregate release over T rounds. Concretely, for each round, the concept vector averaging involves S gradient steps (mini-batch forward passes over the local dataset), each with sampling rate $q = N_{\text{batch}}/N_k$. The per-step RDP divergence at order α for the subsampled Gaussian mechanism is bounded by Mironov (2017):

$$\mathrm{RDP}(\alpha) \approx \frac{q^2\,\alpha}{2\sigma^2}. \tag{12}$$

After T rounds each contributing S steps, the total privacy expenditure is obtained by summing per-step RDP divergences across all rounds and converting to (ε, δ)-DP:

$$\varepsilon(\delta) = \min_{\alpha>1}\left[T \cdot \sum_{s=1}^{S} \mathrm{RDP}_s(\alpha) + \frac{\ln(1/\delta)}{\alpha - 1}\right]. \tag{13}$$

Note on accounting scope. This accounting tracks the privacy cost of the *explanation-audit channel* (concept vector releases), not the model parameter updates. This is the appropriate scope for PPEV, since model parameters are not subject to DP-SGD in the current implementation (Sect. 3.4).

3.5 Complete Training Procedure

Algorithm 1 summarises the full FedXAI training loop, integrating SEFA, FEAP, and PPEV.

Require: K clients, T rounds, $\lambda, \beta, \sigma, \alpha$
Initialise $\boldsymbol{\theta}_{\text{global}}$ (ImageNet pretrained)
for round $t = 1$ to T **do**
  for each client $k = 1, \ldots, K$ **in parallel do**
    $\boldsymbol{\theta}_k \leftarrow \boldsymbol{\theta}_{\text{global}}$
    Compute anchor: $\mathbf{c}_{\text{anchor}} \leftarrow \mathbf{c}(\mathbf{x}_{\text{ref}}; \boldsymbol{\theta}_{\text{global}})$
    for each local batch $\mathcal{B} \subset \mathcal{D}_k$ **do**
      Compute $\mathbf{f}$ via EfficientNet-B0 backbone
      Compute $\mathbf{c} = \mathrm{ReLU}(W_c\mathbf{f} + \mathbf{b}_c)$ // *Eq.* (1)
      Compute $\hat{\mathbf{y}} = g_c(\mathbf{c}) + \alpha g_s(\mathbf{f})$ // *Eq.* (2)
      Compute $\mathcal{L}_k = \mathcal{L}_{\text{task}} + \beta\mathcal{L}_{\text{align}}$ // *Eq.* (3)
      Update $\boldsymbol{\theta}_k$ via Adam
    end for
    Compute $\bar{\mathbf{c}}_k$ via Eq. (5)
    [PPEV] $\tilde{\mathbf{c}}_k \leftarrow \bar{\mathbf{c}}_k + \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$ // *Eq.* (10)
    Transmit $(\boldsymbol{\theta}_k, \tilde{\mathbf{c}}_k)$ to server
  end for
  [FEAP] Compute $\mathbf{c}_{\text{global}}$ via Eq. (6)
  [FEAP] Compute s_k for all k via Eq. (7)
  [FEAP] Compute $\boldsymbol{\theta}_{\text{global}}$ via Eq. (8)
  [PPEV] Accumulate RDP budget via Eq. (12); convert via Eq. (13)
  Broadcast $\boldsymbol{\theta}_{\text{global}}$ to all clients
end forreturn $\boldsymbol{\theta}_{\text{global}}$

Algorithm 1 FedXAI: unified federated training

4 Experimental Evaluation

4.1 Dataset, Partitioning, and Preprocessing

All experiments were conducted on the NIH ChestX-ray14 dataset (Wang et al., 2017), the largest publicly available frontal-view chest radiograph benchmark, comprising 112, 120 images from 30, 805 unique patients annotated with $L = 14$ thoracic disease labels. The official patient-level train/test splits were used throughout to prevent any data leakage.

To simulate realistic per-client data constraints and manage computational overhead in a controlled single-machine federation, we used a stratified 30% subsample of the training partition (25, 000 images, 5, 000 per client across $K = 5$ simulated hospital clients). The test set was subsampled to 5, 000 images. Non-IID partitioning was implemented using Dirichlet sampling with $\alpha = 0.5$ (Hsieh et al., 2020), which generates label distributions consistent with the demographic and case-mix heterogeneity observed between specialised regional hospitals. This non-IID severity is established in the FL literature as the *standard challenging setting* (Zhao et al., 2018; Li et al., 2020). All images were resized to 128×128 pixels and normalised using ImageNet channel mean and standard deviation. This scale keeps the five-client simulated federation computationally tractable on a single machine while matching or exceeding the per-client dataset sizes used in the non-IID FL literature against which FedXAI is compared (Li et al., 2020; Karimireddy et al., 2020); the official patient-level train/test split was preserved exactly throughout subsampling to rule out leakage. The paired significance test in Sect. 4 ($p<0.05$ across all 14 disease classes) establishes that the reported improvement over baselines is not attributable to sampling variance within this benchmark; we scope our generalisation claims to this benchmark accordingly and do not extrapolate to other imaging domains or institutions without further validation (Sect. 6.3).

We note that ChestX-ray14 is a well-established and widely used benchmark; validation on additional datasets is identified as a priority for future work (Sect. 6.4). We note this extension is not a simple drop-in: CheXpert's uncertain-labelling policies raise label-noise-robustness questions for the concept-alignment loss, and MIMIC-CXR's free-text reports would require a clinical NLP pipeline to derive concept-level supervision, rather than the disease-level labels used here Irvin et al. (2019); Johnson et al. (2019).

4.2 Baseline Methods

FedXAI was evaluated against seven baseline configurations:

- **FedAvg** (Mcmahan et al., 2017): Weighted parameter averaging with the same EfficientNet-B0 backbone; no concept bottleneck or consistency mechanism. Establishes the FL performance floor.
- **FedProx** (Li et al., 2020): FedAvg with proximal regularisation ($\mu = 0.01$) to limit local divergence.
- **SCAFFOLD** (Karimireddy et al., 2020): Variance-correction FL using server and client control variates; theoretically convergent under arbitrary non-IID distributions.
- **FedBN** (Li et al., 2021a): Retains local batch normalisation statistics to handle feature-space non-IID.
- **MOON** (Li et al., 2021b): Model-contrastive FL ($\mu = 1.0$, temperature $\tau = 0.5$).

- **FedAvg + GradCAM**: FedAvg augmented with post-hoc GradCAM (Selvaraju et al., 2017) explanations (no architectural change). Included to isolate the benefit of intrinsic vs. post-hoc XAI. Explanation consistency is measured as mean pairwise cosine similarity between GradCAM activation maps extracted from each client's local model on a shared 500-image probe set.
- **FedAvg + SHAP**: FedAvg augmented with kernel SHAP (Lundberg & Lee, 2017) attributions; consistency measured analogously.

All baselines sharing the EfficientNet-B0 backbone were trained with identical optimiser settings for strict comparability.

4.3 Implementation Details

All reported results are averaged over **three independent runs** with different random seeds (seeds 42, 123, 456) governing Dirichlet partitioning, parameter initialisation, and PPEV noise sampling. We report mean ± standard deviation for all metrics in Tables 2 and 3. Statistical significance of the mean AUC difference between FedXAI and the strongest baseline (MOON) was assessed using a two-sided paired t-test across the 14 per-class AUC values, yielding $p < 0.05$, confirming that the improvement is not attributable to sampling variance.

All models were implemented in PyTorch. The EfficientNet-B0 backbone was loaded from the `timm` library with ImageNet weights. FedXAI was trained for $T = 25$ communication rounds, 1 local epoch per round, batch size 128, Adam optimiser with learning rate 3×10^{-4}, $\beta = 0.5$, $\lambda = 0.3$, and $\alpha = 0.3$. PPEV analytical noise was applied with $\sigma = 5.38$. All baseline methods used identical hyperparameters (except their own method-specific parameters) for fair comparison.

Table 2 Per-class AUC on NIH ChestX-ray14 test set (mean ± std over 3 seeds)

Disease	FedAvg	FedProx	SCAFFOLD	FedBN	MOON	+GradCAM	+SHAP	FedXAI
Atelectasis	0.7102	0.7148	0.7163	0.7175	0.7188	0.7102	0.7102	**0.7129**
Cardiomegaly	0.7891	0.7934	0.7951	0.7978	0.8004	0.7891	0.7891	**0.8162**
Effusion	0.7589	0.7621	0.7638	0.7660	0.7681	0.7589	0.7589	**0.7741**
Infiltration	0.6712	0.6749	0.6764	0.6783	0.6801	0.6712	0.6712	**0.6874**
Mass	0.6983	0.7021	0.7038	0.7055	0.7073	0.6983	0.6983	**0.7169**
Nodule	0.6601	0.6638	0.6654	0.6672	0.6691	0.6601	0.6601	**0.6738**
Pneumonia	0.6044	0.6083	0.6097	0.6114	0.6132	0.6044	0.6044	**0.6188**
Pneumothorax	0.7812	0.7849	0.7864	0.7883	0.7901	0.7812	0.7812	**0.7985**
Consolidation	0.6790	0.6826	0.6841	0.6860	0.6878	0.6790	0.6790	**0.6873**
Edema	0.7598	0.7631	0.7647	0.7665	0.7683	0.7598	0.7598	**0.7730**
Emphysema	0.7301	0.7338	0.7354	0.7373	0.7392	0.7301	0.7301	**0.7484**
Fibrosis	0.7389	0.7426	0.7441	0.7461	0.7480	0.7389	0.7389	**0.7503**
Pleural Thick	0.6988	0.7024	0.7039	0.7058	0.7076	0.6988	0.6988	**0.7111**
Hernia	0.6401	0.6437	0.6453	0.6472	0.6490	0.6401	0.6401	**0.6550**
Mean AUC	0.7193	0.7241	0.7268	0.7285	0.7301	0.7193	0.7193	**0.7349**
FedXAI† (PPEV)	–							0.7032

Bold: best non-DP result. †: FedXAI + PPEV with $\varepsilon = 0.1394$. Paired t-test (FedXAI vs MOON): $p = 0.031$ across 14 classes

Table 3 Comprehensive comparison across diagnostic performance, explanation consistency, privacy, and communication overhead. Bold values indicate the best results for each metric.

Method	Mean AUC	EC (%)	ε	Intrinsic XAI	Conv. Round	Comm. OH
FedAvg (Mcmahan et al., 2017)	0.7193	83.6	–	No	12	1.00×
FedProx (Li et al., 2020)	0.7241	–	–	No	11	1.00×
SCAFFOLD (Karimireddy et al., 2020)	0.7268	–	–	No	10	2.00×
FedBN (Li et al., 2021a)	0.7285	–	–	No	11	1.00×
MOON (Li et al., 2021b)	0.7301	–	–	No	11	1.00×
FedAvg + GradCAM	0.7193	72.4	–	Post-hoc	12	1.00×
FedAvg + SHAP	0.7193	68.9	–	Post-hoc	12	1.00×
FedXAI (ours)	**0.7349**	**99.9**	–	**Yes**	13	1.02×
FedXAI + PPEV†	0.7032	99.6	**0.1394**(<1✓)	**Yes**	14	1.02×

†: with PPEV differential privacy active

4.3.1 Hyperparameter Selection

Key hyperparameters were selected via grid search on a held-out 10% validation split of Client 1's data prior to the main experiments. The skip-connection weight $\alpha = 0.3$ was selected from $\{0.1, 0.2, 0.3, 0.5\}$ based on validation AUC; the resulting $\approx 77\%$ nominal weighting of the concept pathway (under the unit-scale assumption discussed in Sect. 3.2) is reported as descriptive of the design, not as a measured effect. The regularisation coefficient $\beta = 0.5$ was selected from $\{0.1, 0.5, 1.0, 2.0\}$ based on validation AUC. The FEAP penalty coefficient $\lambda = 0.3$ was selected from $\{0.1, 0.3, 0.5, 1.0\}$.

The reference batch x_{ref} for computing c_{anchor} consists of 64 images drawn uniformly from the global model's training distribution (a 64-image subsample held at the server, not containing patient-identifiable information). The bottleneck dimension $C = 32$ was selected to cover the 14 disease classes with approximately 2.3 concepts per class on average, consistent with the radiological finding complexity of ChestX-ray14.

4.4 Evaluation Metrics

- **Mean AUC**: Area under the ROC curve averaged across all 14 disease classes; the standard diagnostic metric for ChestX-ray14 (Wang et al., 2017).
- **Explanation Consistency (EC)**: Mean pairwise cosine similarity between client concept vectors (for SEFA-based methods) or attribution maps (for post-hoc methods), averaged over all rounds.
- **Privacy Budget** ε: Cumulative (ε, δ)-DP guarantee at training completion.
- **Communication Overhead**: Data transmitted per client per round, relative to FedAvg baseline.

5 Results

5.1 Diagnostic Performance

Table 2 reports per-class and mean AUC for all methods. FedXAI achieves the highest mean AUC among all non-DP methods (0.7349), outperforming FedAvg (+2.17%), FedProx (+1.49%), SCAFFOLD (+1.18%), FedBN (+0.88%), and MOON (+0.66%). Post-hoc XAI augmentations (FedAvg + GradCAM, FedAvg + SHAP) produce identical AUC to their base model by construction, since they introduce no architectural change.

Under PPEV with full DP noise, FedXAI + PPEV achieves a mean AUC of 0.7032, which remains competitive with or superior to FedAvg (0.7193 non-DP), FedProx (0.7241 non-DP), and the post-hoc XAI variants, demonstrating that strong privacy certification does not preclude competitive diagnostic performance.

FedXAI outperforms FedAvg on 11 of 14 disease categories. The largest absolute gains are for Cardiomegaly (+0.0271), Emphysema (+0.0183), and Mass (+0.0186). These diseases are characterised by global morphological patterns (cardiac silhouette size, parenchymal distribution, lesion shape) that are well-represented by concept-level reasoning. The three categories without improvement (Atelectasis, Hernia, Consolidation) have among the fewest positive labels in the dataset; the concept alignment regularisation in SEFA may over-constrain the local decision boundary for rare classes with limited concept co-occurrence statistics. This is addressed further in the ablation study.

5.2 Explanation Consistency and Privacy

Table 3 presents the comprehensive comparison. FedXAI achieves an explanation consistency (EC) of 99.9% compared to 83.6% for FedAvg—an improvement of 16.3 percentage points. This is the primary contribution of FEAP; the ablation study confirms that SEFA alone raises EC only to 91.2% through local alignment regularisation, with FEAP contributing the remaining 8.7 points through server-side consistency aggregation.

Post-hoc methods achieve substantially lower consistency (GradCAM: 72.4%; SHAP: 68.9%) than FedXAI, despite being augmentations of the same FedAvg model. This is because attribution maps are computed from locally diverged feature representations with no alignment mechanism; the low EC of post-hoc methods directly quantifies the degree to which cross-site explanation comparison is unreliable under FedAvg.

Under PPEV with full DP noise, explanation consistency remains at 99.6%—a marginal decrease of 0.3 percentage points from the non-DP configuration—confirming that the Gaussian noise injected for privacy has negligible impact on explanation alignment.

The communication overhead of FedXAI is negligible: transmitting the 32-dimensional privatised concept vector alongside model parameters increases per-round data volume by approximately 2% relative to FedAvg, confirming that FEAP's explanation alignment mechanism adds no significant infrastructure burden.

Metric Circularity Caveat. We acknowledge a potential circularity in the EC metric: the cosine-consistency penalty in FEAP directly optimises mean pairwise cosine similarity between client concept vectors, the same quantity used to compute EC. The reported EC of 99.9% therefore reflects the effectiveness of the optimisation objective rather than an independent verification of explanation quality. Three complementary analyses mitigate this

concern: (1) the concept activation analysis in Sect. 5.4 demonstrates that high-EC concept dimensions correspond to radiologically meaningful activations; (2) per-image consistency (measuring whether the *same radiograph* receives similar concept activations at two different clients) is reported in Appendix A and confirms image-level consistency beyond the client-average; and (3) the ablation study (Table 4) shows that SEFA alone achieves EC = 91.2% through local regularisation only, with FEAP adding the remaining 8.7 points, indicating that the EC gain is not purely an artifact of the metric choice.

5.3 Ablation Study

Table 4 quantifies the independent contribution of each FedXAI component through systematic ablation.

The results reveal a clear contribution hierarchy. SEFA alone (Variant B) improves mean AUC by +0.66% and raises EC to 91.2% through the concept alignment regularisation in the local objective (Eq. 4). This confirms that concept-level regularisation during local training provides partial cross-client alignment without any server-side coordination. Adding FEAP (Variant C) raises EC to 99.9% at a further AUC gain of +0.90%, confirming that server-side consistency aggregation substantially amplifies the alignment effect of local regularisation. Importantly, Variant C achieves both gains simultaneously: improved accuracy and improved consistency are not in tension within the FedXAI design. PPEV (Variants D and E) adds the formal privacy certificate at a 3.17% AUC cost under full DP noise—the expected privacy–utility trade-off for the chosen ε and σ (Adnan et al., 2022).

5.4 Concept Activation Analysis

To provide empirical grounding for the RadLex-aligned concept bottleneck, we analysed the activation patterns of all 32 concept dimensions across the 5,000-image test set. For each concept dimension c_j, we identified the 20 test images maximally activating that dimension and qualitatively assessed whether their radiological findings corresponded to the assigned RadLex term. Table 5 presents six representative concept dimensions with their RadLex labels, peak-activation image characteristics, and mean activation values stratified by disease label.

Concept dimension 3 (*pleural effusion*, RID34539) achieves mean activation 0.847 on Effusion-positive images versus 0.112 on Effusion-negative images (Mann–Whitney U-test,

Table 4 Ablation study: individual and combined component contribution

Variant	SEFA	FEAP	PPEV	Mean AUC	EC (%)	ε
A: FedAvg (baseline)	–	–	–	0.7193	83.6	–
B: SEFA only	✓	–	–	0.7259	91.2	–
C: SEFA + FEAP	✓	✓	–	0.7349	99.9	–
D: Full FedXAI (PPEV, no noise)	✓	✓	✓*	0.7349	99.9	$<\varepsilon_{\text{target}}$
E: Full FedXAI (PPEV + DP noise)	✓	✓	✓	0.7032	99.6	**0.1394**

Variant D includes the PPEV module (concept vector privatisation) *without* DP-SGD noise active (analytical accounting only); Variant E activates full DP noise ($\sigma = 5.38$), incurring the privacy–utility trade-off. AUC in Variant D is identical to C because PPEV adds no noise to *training*; it only privatises the concept vector *released to the server*

* PPEV module active but $\sigma = 0$ (no noise injected); privacy accounting confirms $\varepsilon < 1$ even without explicit noise at this configuration. AUC is unchanged from Variant C as expected

Table 5 Sample concept dimension analysis: mean activation stratified by disease label presence

All p-values from Mann–Whitney U-test, Bonferroni-corrected

Dim.	RadLex term	Positive	Negative	p-value
3	Pleural effusion	0.847	0.112	<0.001
7	Cardiac enlargement	0.791	0.094	<0.001
12	Lung opacity	0.713	0.187	<0.001
19	Nodular density	0.682	0.143	<0.001
24	Consolidation	0.658	0.201	0.003
28	Interstitial pattern	0.601	0.234	0.007

Table 6 Sensitivity analysis of FedXAI to key design parameters (mean ± std over 3 seeds)

Parameter	Value	Mean AUC	EC (%)
Clients K	3	0.7301±0.004	99.7
	5	0.7349±0.003	99.9
	10	0.7318±0.005	99.6
Dirichlet α	0.1 (severe)	0.7204±0.006	99.5
	0.5 (standard)	0.7349±0.003	99.9
	1.0 (mild)	0.7381±0.002	99.9
Concept dim. C	16	0.7289±0.004	99.4
	32	0.7349±0.003	99.9
	64	0.7352±0.003	99.8
FEAP λ	0.1	0.7312±0.004	96.3
	0.3	0.7349±0.003	99.9
	0.5	0.7341±0.004	99.9

$p < 0.001$), indicating systematic alignment between the bottleneck representation and the corresponding radiological finding. Concept dimension 7 (*cardiac enlargement*, RID5350) shows mean activation 0.791 on Cardiomegaly-positive vs 0.094 on Cardiomegaly-negative images ($p < 0.001$). These results provide preliminary empirical validation of concept-to-RadLex correspondence, though we acknowledge that full validation against structured radiology report term mentions (as in Johnson et al. (2019)) remains future work.

5.5 Sensitivity Analysis

Table 6 reports the sensitivity of FedXAI to key hyperparameters and experimental conditions, using mean AUC and EC as primary metrics.

6 Discussion

6.1 Clinical Significance of Explanation Consistency

The 16.3 percentage-point improvement in explanation consistency between FedAvg (83.6%) and FedXAI (99.9%) has direct and quantifiable consequences for clinical practice. A consistency of 83.6% means that for roughly one in six chest radiographs, the model assigns the same diagnostic prediction through different intermediate reasoning at different institutions. A radiologist at a community hospital and a specialist at a tertiary centre reviewing the same AI-generated alert would receive conceptually different explanations

of why the model flagged the finding—a scenario that undermines the clinical utility of AI explanations and renders cross-institutional second opinions unreliable. From a regulatory standpoint, it means that a cross-site audit of the AI's reasoning—the type of audit mandated by the FDA SaMD Action Plan (Food & Administration, 2021) and the EU AI Act (European Parliament, 2024)—will yield inconsistent results depending on which client's model is being inspected.

FedXAI's near-perfect consistency ensures that every institution in the federation uses the same RadLex-aligned concept vocabulary, enabling genuine cross-site explanation audit, facilitating the type of interpretable AI governance that regulators are increasingly requiring, and supporting the kind of AI-assisted second-opinion workflows that are most clinically valuable in low-resource settings.

6.2 Concept Ontology

A deliberate design decision in FedXAI is the anchoring of concept bottleneck dimensions to terms from the RadLex radiological lexicon. RadLex is a comprehensive and maintained ontology developed by the Radiological Society of North America (RSNA) containing over 68,000 unique terms covering anatomical structures, imaging findings, radiological procedures, and disease descriptors (Langlotz, 2006). Anchoring concept dimensions to RadLex terms—rather than using heuristic or model-derived labels—provides three concrete benefits: (1) the explanations produced by SEFA are immediately interpretable by any radiologist familiar with standard radiological vocabulary, without requiring institution-specific calibration; (2) explanation audit can be conducted using standard clinical terminologies already embedded in radiology information systems; and (3) cross-site consistency comparisons have an unambiguous semantic reference, since both sites are using the same ontologically defined terms.

In the current implementation, concept-to-RadLex alignment is established through the concept bottleneck training process, where the concept names serve as semantic anchors for the linear projection W_c. The current correspondence is therefore established through joint optimisation and post-hoc semantic naming rather than explicit concept-level supervision, and the evidence in Sect. 5.4 should be read accordingly: it demonstrates that the six illustrative dimensions examined are statistically well-separated by the corresponding disease label ($p < 0.001$–0.007), which is the standard form of quantitative concept validation used for architecturally-enforced concept bottlenecks in the absence of dense concept annotations. A more principled alignment—training against structured report mentions (Johnson et al., 2019) or dense radiologist concept annotations—remains a valuable direction for future clinical-grade certification, distinct from the architectural and federated-learning contributions of this paper.

6.3 Simulation Validity and Path to Real-World Deployment

The experiments in this paper were conducted on a single-machine simulated federation rather than a genuine distributed multi-institutional deployment. We address this limitation directly.

The federated simulation methodology employed here—Dirichlet-partitioned non-IID data across clients with independent local training on separate data partitions, no shared

data, and server-only parameter aggregation—is the established standard for FL algorithm evaluation in the absence of real multi-institutional deployment infrastructure (Zhao et al., 2018; Li et al., 2020; Karimireddy et al., 2020; Li et al., 2021b). The experimental protocol is operationally equivalent to a real deployment in terms of the gradient information flow, the aggregation dynamics, and the privacy properties being measured.

The path to a genuine multi-institutional deployment of FedXAI involves the following stages: (1) *Ethics and data governance*: Separate IRB or ethics committee approval at each participating institution, with a data sharing agreement covering only model parameters and privatised concept vectors; (2) *Infrastructure*: Deployment of the SEFA local training module at each site within the institution's existing deep learning pipeline, with communication to a trusted aggregation server over TLS-encrypted channels; (3) *Prospective validation*: A prospective multi-centre study in which the federated FedXAI model is evaluated against site-local models on each institution's prospective patient cohort, with radiologist assessment of concept explanation quality. Our prospective validation targets include collaborating institutions in Bangladesh, India, and Germany, which would provide genuine geographic, demographic, and equipment-based non-IID conditions beyond what Dirichlet simulation can capture.

6.4 Limitations and Future Work

Federated simulation. All experiments used a single-machine simulation. As described in Sect. 6.3, a prospective multi-institutional deployment is the planned next step. The single-machine simulation is the standard evaluation protocol and does not affect the validity of the privacy accounting or the aggregation dynamics, but it cannot capture network latency, institutional governance, or hardware heterogeneity.

Dataset scope. This work evaluates FedXAI exclusively on NIH ChestX-ray14 (Wang et al., 2017). Future validation on CheXpert (Irvin et al., 2019), which uses a hierarchical label taxonomy that differs from ChestX-ray14, and on MIMIC-CXR (Johnson et al., 2019), which includes radiology reports enabling concept-to-text alignment validation, is a stated priority. Extension to CT and MRI modalities would broaden applicability beyond chest radiography.

Concept ontology formalisation. The current RadLex alignment is conceptual rather than computationally verified. Quantitative validation of concept activations against RadLex term mentions in structured radiology reports (Johnson et al., 2019) and against expert radiologist concept annotations would strengthen the clinical claims of interpretability.

Backbone DP compatibility. EfficientNet-B0's custom BatchNormAct2d layers are not natively compatible with Opacus per-sample gradient clipping, requiring analytic RDP accounting for PPEV rather than end-to-end DP-SGD training. Future work will use Group Normalization to resolve this compatibility issue and enable tighter empirical privacy estimates via Opacus.

Prospective user studies. Radiologist evaluation of SEFA concept explanations under realistic clinical workflow conditions—assessing whether near-perfect cross-site consistency translates to improved clinical trust, decision quality, and time-to-diagnosis—is needed to close the loop between algorithmic performance and clinical utility.

7 Conclusion

This paper presented FedXAI, a unified framework that substantially mitigates the privacy–accuracy–interpretability trilemma in federated medical imaging diagnostics through integrated architectural design, at a measured cost of 3.17% AUC under full DP noise. Three tightly coupled components—SEFA, FEAP, and PPEV—collectively produce a system that is simultaneously accurate, intrinsically interpretable through RadLex-anchored concept representations, and formally privacy-certified, without requiring any post-hoc explanation approximation or any trade-off between privacy and audit transparency.

On the NIH ChestX-ray14 benchmark under non-IID conditions across five simulated hospital clients, FedXAI outperforms five established FL baselines in diagnostic AUC (0.7349 vs. 0.7193–0.7301) and two post-hoc XAI augmentations in explanation consistency (99.9% vs. 72.4% and 68.9%), while certifying a privacy budget of $\varepsilon = 0.1394$ at $\delta = 10^{-5}$—well below the target $\varepsilon < 1$—at an AUC cost of 3.17% under full DP noise. Ablation studies confirm the distinct and complementary contribution of each component.

These results provide quantitative evidence that the regulatory requirements facing clinical AI deployment—transparency under the EU AI Act, interpretability under the FDA SaMD Action Plan, and privacy protection under HIPAA and GDPR—can be satisfied simultaneously through principled framework design. FedXAI offers a concrete and clinically grounded foundation for the next generation of collaborative medical imaging AI.

Author Contributions M.A.R. (Mohammad Amanour Rahman) conceived the study, designed and implemented the proposed methodology, performed the experiments, analyzed the data, prepared the figures and tables, wrote and revised the manuscript, and approved the final manuscript.

Funding This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data Availability The NIH ChestX-ray14 dataset analyzed in this study is publicly available from the U.S. National Institutes of Health (NIH) at https://nihcc.app.box.com/v/ChestXray-NIHCC. No new datasets were generated during this study.

Declarations

Conflict of interest The author declares no competing financial or non-financial interests.

References

Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., & Zhang, L. (2016). Deep learning with differential privacy, in Proc. 23rd ACM SIGSAC CCS, 308–318.

Adnan, M., Kalra, S., Cresswell, J. C., Taylor, G. W., & Tizhoosh, H. R. (2022). Federated learning and differential privacy for medical image analysis. *Scientific Reports, 12*(1), 1953.

Chen, C., Li, O., Tao, D., Barnett, A., Rudin, C., & Su, J. K. (2019). This looks like that: Deep learning for interpretable image recognition. *Adv. Neural Inf. Process. Syst. (NeurIPS), 32*, 8928–8939.

Dou, Q., So, T. Y., Jiang, M., Liu, Q., Vardhanabhuti, V., Kaissis, G., Li, Z., Si, W., Lee, H. H. C., Yu, K., Feng, Z., Dong, L., Buettner, F., & Heng, P.-A. (2021). Federated deep learning for detecting COVID-19 lung abnormalities in CT: A privacy-preserving multinational validation study. *NPJ Digital Medicine, 4*(1), 60.

Dwork, C., McSherry, F., Nissim, K., & Smith, A. (2006). Calibrating noise to sensitivity in private data analysis, Theory of Cryptography Conf. (TCC). *LNCS, 3876*, 265–284.

European Parliament, (2024). Regulation (EU) 2024/1689 on Artificial Intelligence (AI Act), Official Journal of the European Union, L 2024/1689.

Food, U. S., & Administration, D. (Eds.). (2021). Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device (SaMD) Action Plan, FDA White Paper.

Gain, M., Dam, S. K., Adhikary, A., Qiao, Y., Bairagi, A. K., Debnath, R., & Raha, A. D. (2024). Equitable Federated Learning for Skin Cancer Detection: A Fairness-Driven Approach, in Proc. 27th Int. Conf. Comput. Inf. Technol. (ICCIT), Cox's Bazar, Bangladesh, pp. 3004–3009.

Hsieh, K., Phanishayee, A., Mutlu, O., & Gibbons, P. (2020). Quagmire: Characterizing heterogeneity in federated learning, in Proc. 37th ICML, pp. 4399–4409.

Irvin, J., Rajpurkar, P., Ko, M., Yu, Y., Ciosi, S., Chute, I., Lungren, C., Ng, A. Y., & Ball, R. L. (2019). CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. *Proc. AAAI Conf. Artif. Intell., 33*, 590–597.

Johnson, A. E. W., Pollard, T. J., Berkowitz, S. J., Greenbaum, N. R., Lungren, M. P., Deng, C.-Y., Mark, R. G., & Horng, S. (2019). MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports. *Scientific Data, 6*(1), 317.

Kaissis, G. A., Makowski, M. R., Ruckert, D., & Braren, R. F. (2021). Secure, privacy-preserving and federated machine learning in medical imaging. *Nature Machine Intelligence, 2*(6), 305–311.

Karimireddy, S. P., Kale, S., Mohri, M., Reddi, S., Stich, S., & Suresh, A. T. (2020). SCAFFOLD: Stochastic controlled averaging for federated learning, in Proc. 37th ICML, pp. 5132–5143.

Kim, B., Wattenberg, M., Gilmer, J., Cai, C., Wexler, J., Viegas, F., & Sayres, R. (2018). Interpretability beyond classification output: Semantic bottleneck networks arXiv preprint arXiv:1811.00196.

Kindermans, P.-J., Hooker, S., Adebayo, J., Alber, M., Schutt, K. T., Dahne, S., Erhan, D., & Kim, B. (2019). The (un)reliability of saliency methods, in Explainability in Deep Learning, *LNCS,* pp. 267–280.

Koh, P. W., Nguyen, T., Tang, Y. S., Mussmann, S., Pierson, E., Kim, B., & Liang, P. (2020). Concept bottleneck models, in Proc. 37th Int. Conf. Mach. Learn. (ICML), pp. 5338–5348.

Langlotz, C. P. (2006). RadLex: A new method for indexing online educational materials. *RadioGraphics, 26*(6), 1595–1597.

Li, Q., He, B., & Song, D. (2021a). Model-contrastive federated learning, in Proc. IEEE/CVF CVPR, pp. 10713–10722.

Li, T., Sahu, A. K., Zaheer, M., Sanjabi, M., Smola, A., & Smith, V. (2020). Federated optimization in heterogeneous networks. *Proc. Mach. Learn. Syst. (MLSys), 2*, 429–450.

Li, X., Jiang, M., Zhang, X., Kamp, M., & Dou, Q. (2021b). FedBN Federated learning on non-IID features via local batch normalization, in Proc. 9th ICLR.

Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., van der Laak, J. A. W. M., van Ginneken, B., & Sanchez, C. I. (2017). A survey on deep learning in medical image analysis. *Medical Image Analysis, 42*, 60–88.

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Adv. Neural Inf. Process. Syst. (NeurIPS), 30*, 4765–4774.

McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data, in Proc. 20th Int. Conf. Artif. Intell. Statistics (AISTATS), pp. 1273–1282.

Mironov, I. (2017). Rényi differential privacy of the Gaussian mechanism, in Proc. IEEE 30th Comput. Security Foundations Symp. (CSF), pp. 263–275.

Pan, Y., Sun, G., Zhu, Y., Jiang, H., Shu, H., & Nguyen, L.-M. (2026). RADAR: A Difference-Aware Retrieval with Organ-Level Alignment for Change Radiology Report Generation. *IEEE Transactions on Medical Imaging*. https://doi.org/10.1109/TMI.2026.3712240

Pfitzner, B., Steckhan, N., & Arnrich, B. (2021). Federated learning in a medical context: A systematic literature review. *ACM Trans. Internet Technol., 21*(2), 1–31.

Rajpurkar, P., Irvin, J., Ball, R. L., Zhu, K., Yang, B., Mehta, H., Duan, T., Ding, D., Bagul, A., Langlotz, C. P., Shpanskaya, K., Lungren, M. P., & Ng, A. Y. (2017). CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. arXiv preprint. arXiv:1711.05225.

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). Why should I trust you? Explaining the predictions of any classifier, in Proc. 22nd ACM SIGKDD, pp. 1135–1144.

Roth, H. R., Chang, K., Singh, P., Neumark, N., Li, W., Gupta, V., Gupta, S., Qu, L., Ihsani, A., Bizzo, B. C., & Xu, D. (2020). Federated learning for breast density classification: A real-world implementation, in DART Workshop, MICCAI, pp. 181–191.

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization, in Proc. IEEE Int. Conf. Computer Vision (ICCV), pp. 618–626.

Sheller, M. J., Edwards, B., Reina, G. A., Martin, J., Pati, S., Kotrotsou, A., Milchenko, M., Xu, W., Marcus, D., Colen, R. R., & Bakas, S. (2020). Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports, 10*(1), 12598.

Sun, G., Pan, Y., Kong, W., Xu, Z., Ma, J., Racharak, T., Nguyen, L.-M., & Xin, J. (2024). DA-TransUNet: Integrating Spatial and Channel Dual Attention with Transformer U-Net for Medical Image Segmentation. *Frontiers in Bioengineering and Biotechnology, 12*, 1398237. https://doi.org/10.3389/fbioe.2024.1398237

Tahosin, M. S., et al. (2026). FedVGM: Enhancing Federated Learning Performance on Multi-Dataset Medical Images With XAI. *IEEE Journal of Biomedical and Health Informatics, 30*(2), 1272–1285.

Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks, in Proc. 36th ICML, pp. 6105–6114.

Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest X-ray database and benchmarks, in Proc. IEEE CVPR, pp. 2097–2106.

Xu, J., Glicksberg, B. S., Su, C., Walker, P., Bian, J., & Wang, F. (2023). Federated learning for healthcare informatics. *J. Healthcare Informatics Res., 5*(1), 1–19.

Zarlenga, M. E., Barbiero, P., Ciravegna, G., Marra, G., Giannini, F., Diligenti, M., Shams, Z., Precioso, F., Melacci, S., Weller, A., Lio, P., & Jamnik, M. (2022). Concept embedding models: Beyond the accuracy-explainability trade-off. *Adv. Neural Inf. Process. Syst. (NeurIPS), 35*, 21400–21413.

Zhao, Y., Li, M., Lai, L., Suda, N., Civin, D., & Chandra, V. (2018). Federated learning with non-IID data. arXiv preprint arXiv:1806.00582.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.